5
Module 47

Science and Health

WIND MOVEMENT

A DepEd-BEAM Distance Learning Program supported by the Australian Agency for International Development

To the Learner

You have learned that land is heated faster than water. Do you think this unequal heating of land and water will produce good effects? How? Let's find out by going through this module.

Let's Learn This

Describe the effect of unequal heating of land and water.

Let's Try This

Encircle the letter that has the correct answer.

1. What makes you cool during the day on the beach?

   a.) mountain breeze c.) valley breeze
   b.) sea breeze d.) land breeze

2. What makes you cool at night on the beach?

   a.) mountain breeze c.) valley breeze
   b.) sea breeze d.) land breeze

3. What does unequal heating of land and water produce?

   a.) wind b.) rain c.) tiles d.) surge

4. Which describes unequal heating?

I. Land heats faster than water.
II. Water takes longer time to cool.
III. Land and water heats up at the same time.
IV. Land cools faster than water.

a.) I, II & III b.) I, II & IV c.) III, IV & II d.) IV, III & I

5. Which situation show wind movement?

I. It is very hot today.
II. It is very cold today.
III. It's refreshing when I feel the sea breeze at the beach.
IV. Its natural for air to go up when heated.

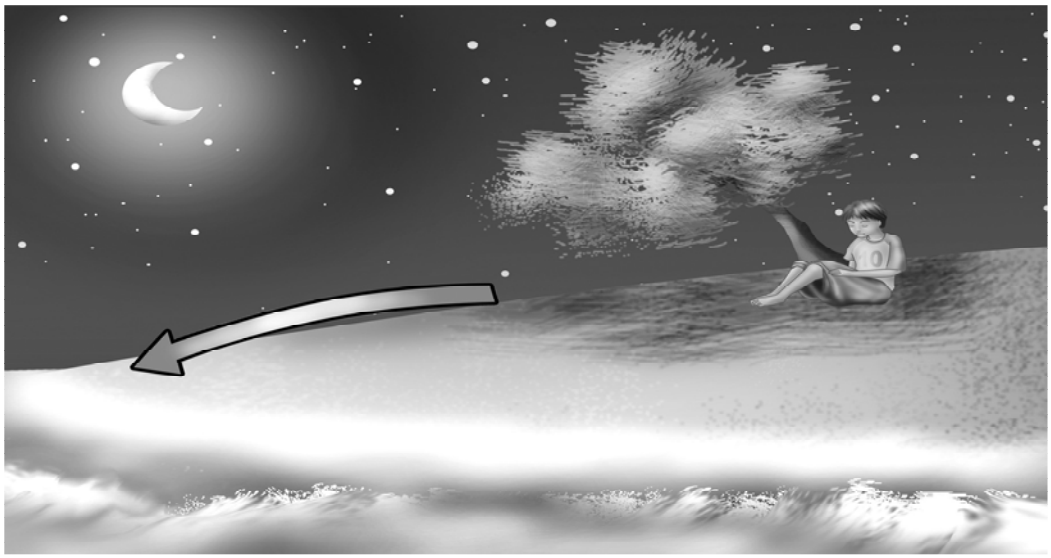
a.) I only
b.) I & II only
c.) III only
d.) III & IV only

Let's Study This

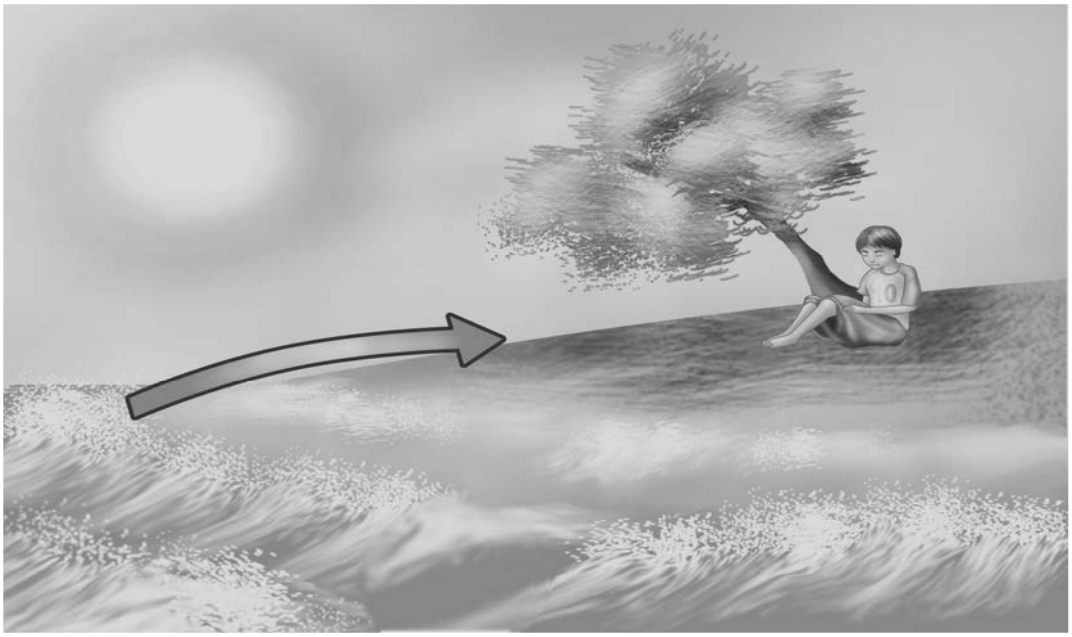
What is the difference between air and wind?

How does wind came about?

Study the illustration below.



LAND BREEZE



SEA BREEZE

During the day, the land and the sea are heated by the sun. Land is heated faster so the air over land is hot and rises. The air over water is cooled and sinks. The weight of air over land and air over the sea water are not equal. Warm air has low pressure because it is less heavy so it goes up. Cool air has high pressure because its more heavy so it goes down. The space occupied by warm air will be replaced and occupied by cool air, thus wind is produced. Wind is the result of air moving due to unequal or differential pressures. This explains the movement of air called **BREEZE**.

At daytime the warm light air over land rises and cool air above the sea replaces the warm air and produces a breeze called **SEA BREEZE**.

At night, the land cool faster so air above land replaced the warm air over sea that rises. Forming a breeze called LAND BREEZE.

Let's Do This

Rearrange the letter to form words.

1. The movement of air. DIWN
2. Breeze that occurs during daytime. AES BEEZRE
3. Breeze that occurs during nighttime. DALN REEZBE
4. Produces wind. QUALUNE INGTEAH

Let's Do More

Draw an arrow to illustrate the movement of air and identify what kind of breeze.

1. _____ breeze

2. _____ breeze

3.) In what direction does sea breeze flow?

4.) In what direction does land breeze flow?

Let's Remember This

- **Unequal heating of land and water produces air to move or wind.**
- **Land breeze happens at night.**
- **Sea breeze occurs at daytime.**

Let's Test Ourselves

Encircle the letter that has the correct answer.

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Science Fact File

Difference in the heating and cooling rates of land and water cause the breezes called the "Daily Monsoons".

Mountain regions also have their "Daily Monsoons" called mountain breeze and the valley breeze.

Answer Key

Let's Try This

1.) b
2.) d
3.) a
4.) b
5.) c

Let's Do This

1.) Wind
2.) Sea Breeze
3.) Land Breeze
4.) Unequal Breeze

Let's Do More

1.) Sea Breeze =
2.) Land Breeze =
3.) Sea to Land
4.) Land to Sea

Let's Test Ourselves

1.) b
2.) d
3.) a
4.) b
5.) c